

Payman Sharafianardakani

Robotics Research Engineer | Ph.D. Candidate | Full-Stack Robotician

Louisville, KY • **Work Authorization: EAD Holder (Green Card Pending) – No Sponsorship Required**

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PROFESSIONAL SUMMARY

Robotics engineer with 8 years of research and system-integration experience spanning physical human-robot interaction, safety-critical shared control, sensor fusion, and deep learning. Lead engineer on an NSF-funded clinical mobile manipulator, responsible for the stack end to end: motor-controller PCBs and SLA-printed sensor hardware, ROS control and perception nodes, MPC-CBF safety filters, a browser teleoperation front end, and the IRB human-subject trials that validated all of it. Five peer-reviewed journal papers and hospital deployment with real patients and nurses.

TECHNICAL SKILLS

Languages: Python, C++, MATLAB, TypeScript/JavaScript, Bash

Robotics: ROS1 (Kinetic/Melodic/Noetic), ROS2 (Humble), Gazebo, RViz, Motion & Path Planning, MoveIt, ROS Navigation Stack (AMCL, costmaps), TF2, Kinova Kortex, Mecanum omnidirectional kinematics, URDF, NVIDIA Isaac Sim

Control & Optimization: PID/PD Loop Tuning (classical control), MPC, Control Barrier Functions (CBF-QP, MPC-CBF), OSQP, CasADi, Admittance/Impedance Control, Neuroadaptive Control, Lyapunov Stability Analysis, Genetic Algorithms (PyGAD), Real-Time Control (50–333 Hz loops)

Estimation & Sensor Fusion: Kalman Filtering (Singer motion model), multi-modal tactile + force/torque fusion, LiDAR obstacle sectoring, network-delay estimation

Deep Learning: PyTorch, OpenCV, TensorFlow, Keras, Scikit-learn, ONNX, CUDA, Semantic Segmentation (zero-shot, FastSAM), Temporal Convolutional Networks, BiLSTM, CNN-GRU, MLP, PointNet++, FastSAM, Contact-GraspNet, Reinforcement Learning

Embedded & Systems: Linux, real-time embedded firmware (RTOS), EtherCAT motor-control chains, CAN, SPI, I2C, UART, Arduino, Teensy, STM32 Nucleo, Sabertooth motor drivers, Jetson, VersaLogic single-board computers, analog signal conditioning, socket programming

Web & Infrastructure: distributed system architecture, CI/CD (GitHub Actions), Next.js (App Router), React, TypeScript, Tailwind CSS, static-export SPA, WebSockets, rosbridge, Cloudflare Tunnel + Access (SSO), CBOR/JPEG stream compression, Docker, Git

Hardware & Prototyping: SolidWorks (DFM/assembly), FDM and SLA 3D printing, PCB design/debug/soldering, FSR tactile arrays (FlexiForce A201), ATI Axia 80 F/T sensors, multi-channel data acquisition (DAQ), RealSense RGB-D, 3D LiDAR

Research Methods: IRB protocol authoring, within-subjects/counterbalanced experiment design, linear mixed-effects models in R, Technology Acceptance Model (TAM)

PROFESSIONAL EXPERIENCE

Lead Robotics Engineer (Ph.D. Research Assistant)

Sep. 2022 – Present

Louisville Automation & Robotics Research Institute (LARRI), University of Louisville

Louisville, KY

- Lead engineer on ARNA, the Adaptive Robotic Nursing Assistant funded by the NSF Future of Work program (#FW-HTF-RM): an omnidirectional mobile manipulator – a 7-DOF Kinova Gen3 arm on a Mecanum base – deployed on an active hospital unit. Responsible for its control, perception, embedded firmware, web infrastructure, and clinical trials.
- Designed and built ARNA's complete remote-operation stack – network-aware safety filtering, secure browser transport, and one-click autonomous grasping – and validated it with **30 genuinely remote operators** across multiple U.S. cities, every command crossing the public internet.
 - * Architected a five-layer shared-control stack for remote teleoperation that holds hard safety guarantees when *both* the operator and the connection are unreliable, under one invariant: adaptive layers may only make the robot more cautious, and barrier constraints are never relaxed. A 100 Hz arm MPC-CBF gave **zero boundary violations in 60 runs** at jerk cost **200–440× lower** than raw operator input; a 50 Hz OSQP base filter on LiDAR sectors gave zero margin violations; a watchdog scaled margins as the link degraded, with a 3.0 s dwell before recovery.
 - * Built a **TypeScript/Next.js** teleoperation interface an operator opens as a URL – no client install, no VPN, and **no inbound port on the robot**, via an outbound Cloudflare Tunnel with identity-based Access. Compiled it to a **fully static bundle** with no Node runtime on the robot. Split the session across **three isolated rosbridge instances** so camera bursts cannot head-of-line-block a joystick or stop command, and cut per-frame data volume **~35–40×** with source-side JPEG encoding and CBOR subscriptions.
 - * Reduced remote manipulation to a single click on a single pixel: FastSAM segments the clicked object on CPU, and Contact-GraspNet with a PointNet++ encoder proposes ranked SE(3) grasps. Deployed it on a **4 GB laptop GPU** by cropping inference to the object bounding region ($\sim 10\times$ input reduction) and warm-loading weights once at node start. Re-ranked candidates **by reachability rather than network confidence** – the binding constraint in practice.

- Delivered a sensorized handlebar and real-time adverse-event detector end to end – mechanical design, electronics, ML, and human trials. Built it around **eight FlexiForce A201 FSRs** and a custom Arduino signal-conditioning board publishing at **122 Hz**, fused with 6-axis F/T through a **Singer-model Kalman filter**. Benchmarked TCN vs. BiLSTM vs. CNN-GRU under subject-wise cross-validation, then deployed the TCN to ONNX onboard the robot – **no cloud** – re-evaluating every **8.2 ms**. On **10 unseen participants: 85.0% detection at 2.05 s mean latency**.
- Learned the joystick-to-robot teleoperation mapping as a black box, with no kinematic model of either device: two networks in parallel, one searched offline by a genetic algorithm to minimize task time, one trained online by SGD against measured jerk. In a 20-subject IRB study, **linear jerk halved** and **angular jerk fell $\sim 3\times$ ($p < .001$)**, **with no increase in task-completion time**.
- Designed a model-free neuroadaptive admittance controller that learns ARNA's unmodelled dynamics online, with update laws derived from a Lyapunov analysis, running at 333 Hz against a 125 Hz admittance model. Against a tuned PD baseline with 10 users: tracking accuracy improved up to **32%** and motion jerk fell up to **52%**. Re-tuned with **63 nursing students**: velocity tracking error **0.0157 $\rightarrow$ 0.0105 m/s**, mean actuator torque **3.66 $\rightarrow$ 1.79 Nm**.
- Designed and led an acceptability trial on an active unit at University of Louisville Hospital with **10 patients and 5 nurses** – real patients, not a lab surrogate. Paired objective task-time logging with Technology Acceptance Model instruments, analyzed with linear mixed-effects models in R: the joystick was significantly easier to use and more likely to be adopted than the tablet ($p \approx 0.02$ – 0.03) *despite statistically similar task times*.
- Repaired and upgraded ARNA's motor-controller PCBs, brought non-functional sonar and bump sensors back online and into ROS, repurposed the LiDAR for navigation, and rebuilt onboard wiring across the EtherCAT motor chain, Sabertooth motor drivers, Teensy microcontrollers, battery/inverter bank, and instrumentation board. Worked across the robot's serial and fieldbus layers – **EtherCAT, CAN, SPI, I2C, and UART** – to bring sensors and drives back under ROS control.
- Mentored **one M.S. student and six undergraduate researchers**, and authored the lab manuals and Standard Operating Procedures for ARNA that new researchers onboard from.

Mechatronics Engineer (M.Sc. Research Assistant)

Nov. 2017 – Sep. 2021

University of Tehran – Human-Robot Interaction Lab

Tehran, Iran

- Engineered a sound-source localization algorithm for an interactive social robot used in autism therapy, achieving a **70% improvement** in audio filtering and a **40% increase** in gesture detection accuracy.
- Designed the “Baby Tank” open-source mobile platform, developing its navigation/path-planning algorithms and managing low-level microcontroller integration for active damping.
- Built a cross-platform Activity Reminder GUI in **C++ and Qt Creator**, focusing on accessibility standards.

EDUCATION

Ph.D. Candidate, Electrical & Robotics Engineering , University of Louisville	Louisville, KY	Sep. 2022 – Dec. 2026 (expected)
M.S. in Mechatronics Engineering , University of Tehran	Tehran, Iran	Nov. 2017 – Sep. 2021
B.Sc. in Electrical Power Engineering , Islamic Azad University	Varamin, Iran	Sep. 2012 – Aug. 2017

SELECTED PUBLICATIONS

Full list: paymansharafian.github.io/publications • [Google Scholar](#)

- P. Sharafianardakani**, M. A. Hanafy, I. Kondaurova, A. Ashary, M. M. Rayguru, D. O. Popa. “Adaptive User Interface With Parallel Neural Networks for Robot Teleoperation.” *IEEE Robotics and Automation Letters*, 10(2):963–970, 2025.
- P. Sharafianardakani**, S. Abubakar, S. K. Das, S. Surupa, M. A. Hanafy, M. Rayguru, D. O. Popa. “Evaluation of a Neuroadaptive Admittance Controller for Ambulation through Physical Human–Robot Interaction.” *Intelligent Service Robotics*, 18(7):1355–1368, 2025.
- P. SharafianArdakani**, I. Kondaurova, A. Ashary, N. Zhang, M. C. Logsdon, M. Walker, D. O. Popa. “Hospital Trial Results for the Acceptability of an Adaptive Robot Nursing Assistant.” *IEEE CASE*, pp. 2043–2049, 2025.
- I. Kondaurova, **P. Sharafian**, R. Mitra, M. M. Rayguru, *et al.*, D. O. Popa. “Acceptance of an Adaptive Robotic Nursing Assistant for Ambulation Tasks.” *Robotics (MDPI)*, 14(9):121, 2025.
- M. C. Logsdon, I. Kondaurova, N. Zhang, *et al.* (incl. **P. SharafianArdakani**), D. O. Popa. “Perceived Usefulness of Robotic Technology for Patient Fall Prevention.” *Workplace Health & Safety*, 72(12):542–549, 2024.
- P. Sharafianardakani et al.** “Tactile Handlebar and Deep Learning for Safe Interaction with a Robot Nursing Assistant.” *IEEE Transactions on Medical Robotics and Bionics* – **under review**.

LEADERSHIP & AWARDS

Journal Reviewer, IEEE Robotics and Automation Letters (RA-L), 2024–2025 • **Conference Reviewer**, ACM/SIGAPP SAC, 2024 • **STEM Mentor**, RAMPS Program – 30+ students, 2023–Present

Dissertation Completion Award (full tuition + stipend), University of Louisville, 2026 • **IEEE RAS Travel Support Award** (ICRA 2025) • **Graduate Student Travel Award**, 2025